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Wind Turbine

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NOTATION	DESCRIPTION	UNIT
<i>Geometry</i>		
L	Total tower height	m
H_w	Submerged height	m
H_a	Height above waterline	m
D_o	Outer diameter	m
t	Wall thickness	m
<i>Structural properties</i>		
E	Young's modulus	Pa
ρ	Steel density	kg/m
A	Cross-sectional area	m^2
I	Second moment of area	m^4
EI	Flexural stiffness	Nm^2
ξ	Damping ratio	-
α, β	Rayleigh damping coefficients	-
γ		-
<i>Dynamic quantities</i>		
ω		
f		
$\ddot{u}$	Second time derivative of displacement	
u''	Second spatial derivative of displacement	
$H(\omega)$		_*
$S(\omega)$		*
σ	Standard deviation	"varies"
<i>Wave and soil</i>		
H_s		m
T_p	Peak wave period	s
G		
ρ_s		
σ_u	Standard deviation of wind turbulence	
L_u		
z	height	m
ρ_w	seawater density	
Ca	added mass coefficient	-
Cm	inertia coefficient	-
Cd	drag coefficient	-
η		
m_{add}	added mass per unit length	kg/m
c_s		
k		

* In this context this is the unit but FRF can be used in other contexts where other units are used

1. Introduction

1.1 Problem description

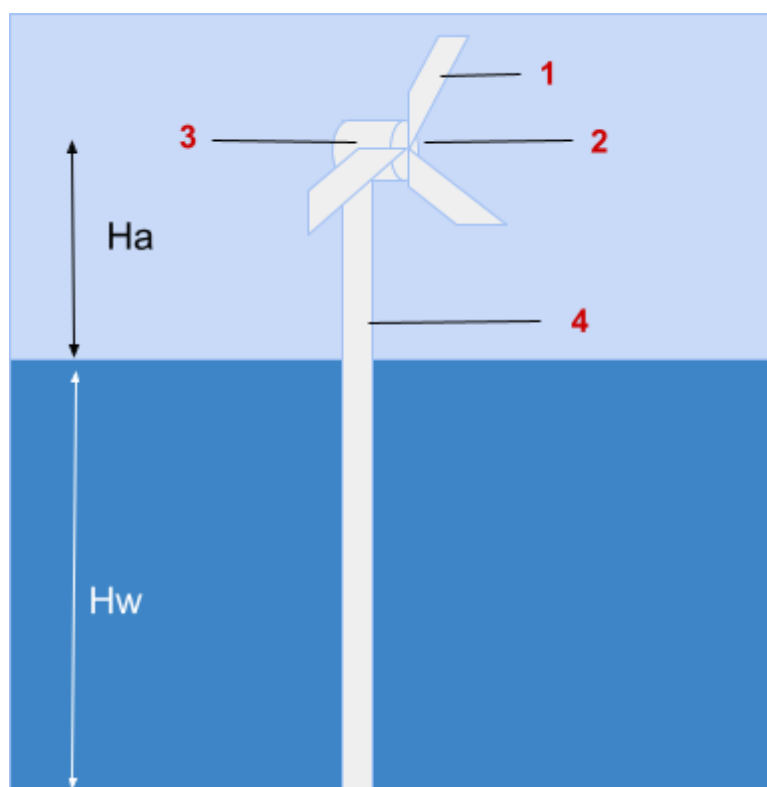
Analyzing loads from natural flows with somewhat irregular behaviour and the effects they have on a structure whose main purpose is to extract energy from these natural forces which creates an interesting and immediate relation to the dynamic response from irregular loads. In this case, an offshore wind-turbine, purposefully suffering from the wind- and water elements. The water and wind pushes as periodic lateral loads that can cause resonance while the wind also turns the rotors in the nacelle, creating a frequency (and a third) that needs to be taken into account regarding the natural frequencies of the structure.

By describing the different kinds of wave and wind spectrums, we are introduced to how these irregular behaviors are accounted for and it will become clear how the spectrum then should be handled to translate into probabilities and dimensioning of real structures. In other terms, the dynamic response is assessed in the frequency domain using power spectral density methods, combining wave and wind load spectra with the structural frequency response function.

Furthermore, an investigation of the reaction, in the (soft) seabed subsoil, to an earthquake, and the risk of resonance there can be found in Report ____, found at page xx in this pdf.

1.3 Language clarification and notations

The wind turbine consists of many mechanical parts which will be referred to without any further descriptions in the report. Some of this specified language will be introduced here to make the report available to a general scholar of a structural dynamic field who is interested in waveloads, frequencies and resonance.



Nacelle being the large powerhouse at the top of the tower, housing the energy generating components. The hub is the part installed at the far end of the nacelle with rotors attached to it. The rotors collect the wind load and converge to RPM.

- 1. Rotor
 - 2. Hub
 - 3. Nacelle
 - 4. Tower length L
- H_a
 H_w

1.2 Assumptions and simplifications

The wind turbine is assumed to have the same structure as an euler-bernoulli beam with a consistent cross section throughout the tower, when wind turbines in fact have a more narrow cross-section at the top than at the bottom.

Assuming that the pile is in a perfectly clamped condition at the seabed and thereby fixed against rotation, does not represent reality where the soil-pile interaction introduces some flexibility. This can be modelled using a winkler foundation with independent springs proportional to local displacement. Some geotechnical investigation would be appropriate to gain information about how the sublayers actually would react to the vibration, weight and pressure from the structure. Some further reading about the sublayers can be found in report XX though with a focus on seismic excitation, not actual foundation stability. Regarding the subject investigated in this report, the simplifications have an impact on the structural stiffness, overestimating it, and thereby slightly overestimating the natural frequencies.

The simplification of the cross section being constant should be adapted and will be commented on later in the report if there is an opportunity to analyze this further. For now we can conclude that it has an impact both on deformations, stiffness and therefore natural frequencies.

The self-weight of the structure and the buoyancy force acting on the submerged section are neglected in the dynamic analysis. Self-weight introduces axial compression which reduces the effective lateral stiffness through geometric nonlinearity. Buoyancy reduced the effective submerged weight. Both effects are considered secondary for a slender offshore pile under lateral wave and wind loading and are commonly neglected in preliminary dynamics analyses of this type.

When a submerged structure accelerates through water it must also accelerate the surrounding fluid, requiring additional force beyond that needed to accelerate the structure itself. This effect is captured by the added mass coefficient C_a , defined such that the total inertia force per unit length on a cylinder is:

$$F_{inertia} = \rho_w \cdot C_m \cdot \left(\pi \frac{D^2}{4}\right) \cdot \ddot{u}_{fluid}$$

where $C_m = 1 + C_a$ is the inertia coefficient. For a circular cylinder $C_a=1$, giving $C_m=2$. The added mass per unit length is therefore:

$$m_{add} = \rho_w C_a \cdot \pi \frac{D^2}{4}$$

In the analysis C_a is included in the Morison force calculation through $C_m=2$, correctly accounting for the added inertia in the forcing. However the added mass is not included in the structural mass matrix M . This is a simplification, in a more complete model the added mass would be added to the diagonal mass terms of the submerged elements, slightly increasing the effective mass and therefore lowering the natural frequencies of the submerged portion of the structure.

2. Structural Model

2.1 Geometry and Properties

The wind turbine can be simplified to an Euler-Bernoulli cantilever beam. The boundary conditions are thereby described by viewing the seabed as a clamped edge containing the maximum moment, while the maximum deformation is located at the top, free end. The properties describing the wind turbine can be found in *Table 1*.

Properties	Values
L	144 m
H_w	36 m
H_a	108 m
E	210 GPa
ρ	7850 kg/m ³

The cross-section is a steel thin-walled cylinder assumed to be uniform. The properties of the cross-section depend on the diameter where the area and area moment of inertia is formulated as such:

$$A = \frac{\pi}{4} (D_o^4 - D_i^4)$$

Eq 1

$$I = \frac{\pi}{64} (D_o^4 - D_i^4)$$

Eq 2

Further dependencies, ratios and tuning of the diameter is described in section 3.2 *Diameter Selection*. Temporary definitions with placeholder values for the matrix assemblies are described in *Table 2*.

Table 2, Cross-section properties

Properties	Values
D_o	5 m
t	0.08 m
D_i	m
A	m ²

I m^4

2.2 FEM

Since the continuous beam has infinitely many degrees of freedom, we discretize it into twelve elements of equal length, where three elements are submerged and nine elements are above the water surface. Since the wind turbine is handled as an Euler-Bernoulli beam, each element has 4 degrees of freedom and each node has two degrees of freedom, represented by the lateral and rotation displacement (w, θ).

The deflection in each element is approximated by summing all the shape functions, as such:

$$w(x) = \sum_{i=1}^4 N_i(x) \cdot u_i$$

Eq 3

with the shape functions defined in a classical way that is applicable for structures of all forms..

$$N_1 = 1 - 3\xi^2 + 2\xi^3$$

$$N_2 = l_e (\xi - 2\xi^2 + \xi^3)$$

$$N_3 = 3\xi^2 - 2\xi^3$$

$$N_4 = l_e (-2\xi^2 + \xi^3)$$

where $\xi = \frac{x}{l_e} \in [0, 1]$

The element stiffness matrix is then constructed with the shape functions with the second derivatives that are connected to the bending strain which depends on the curvature w'' .

$$K_e = EI \int_0^{l_e} N''(x)^T N''(x) dx$$

Eq 4

$$K_e = \frac{EI}{l_e^3} \begin{vmatrix} 12 & 6l_e & -12 & 6l_e \\ -6l_e & 4l_e^2 & -6l_e & 2l_e^2 \\ -12 & -6l_e & 12 & -6l_e \\ 6l_e & 2l_e^2 & -6l_e & 4l_e^2 \end{vmatrix}$$

Eq 5

The element mass matrix is derived from the kinetic energy. It distributes the element's mass according to the shape functions rather ascribing it to the nodes:

$$M_e = \rho A \int_0^{l_e} N(x)^T N(x) dx$$

Eq 6

The global stiffness and mass matrices are then assembled and treated as consistent. by adding each element's stiffness and mass matrices as shown in *Figure 2*.

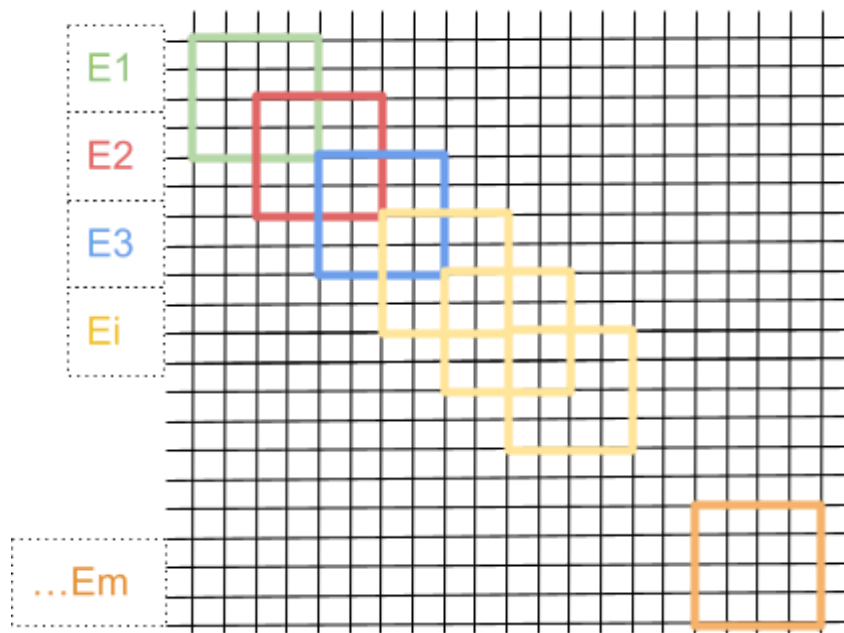


Figure 2. How element matrices are assembled in an additive manner from element 1 (E1) to the last element (Em) where m represents the number of elements

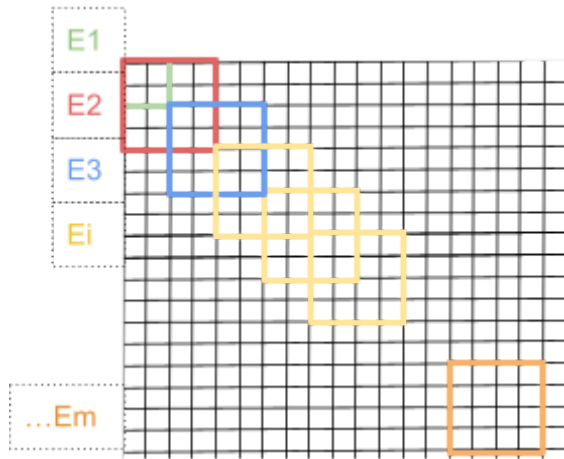
Each element consists of 4 degrees of freedom (DOFs) and there are, in this case, 12 elements. Each element shares the end node resulting in global matrices in the size [26x26].

Nodes=elements+1

ndof=dof*nodes

Size K&M=[ndof x ndof]

Because of the clamped foundation, the rotation and displacement DOFs from the first element are removed from the global matrices. $w(0)=0$, $\theta(0)=0$, zero displacement and zero rotation at the base. Leaving matrices of the size [24x24]



2.3 Sanity check

To ensure that the assembly is programmed in a correct manner, a distributed load of 1000 N/m is applied over the full height and the resulting tip displacement and moment at the seabed are compared to the classically derived expressions for a uniformly loaded cantilever beam.

$$u_{tip} = \frac{qL^4}{8EI}$$

Eq 2

$$M_{base} = \frac{qL^2}{2}$$

Eq 3

Quantity	Analytical (Eq 2, 3)	FEM	Discrepancy
Tip displacement	0.0824 m	0.0824 m	0 %
Moment at seabed	10.368 MNm	10.378 MNm	0.1%

Comparing the results, the displacement at the tip is exact and the mudline moment shows merely a 0.1% discrepancy, confirming that the model works and the code is correctly implemented. Stiffness matrix and assembly are correct.

Before moving forward to the natural frequency tuning, the tip nacelle mass, $m_{nacelle}$, has to be introduced. It is done so as a point mass at the top node in the mass-matrix, M , as such:

$$M_{26,26} += M_{nacelle}$$

This shifts the natural frequency downward significantly, making it crucial to add before tuning the diameter as seen in a simple SDOF system where:

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

A sanity check for the mass matrix, the translational diagonal entries sum to the total structural mass when added across all elements i.e total mass conservation.

$$m_{total} = \rho A \cdot L$$

3. Natural Frequency Tuning

3.1 Design Frequency Window

When dimensioning a wind turbine the periodic excitation from the rotor has to be taken into account. A rotor spinning at a frequency Ω produces a primary excitation at the so-called 1P frequency, corresponding to one pulse per revolution. For a three-bladed rotor, an additional excitation occurs at three times this frequency (3P) as each blade passage induces a load impulse on the tower. This entails that to avoid resonance the first natural frequency of the structure must be a certain frequency window, above 1P and below 3P.

Three design zones are recognized as described in *Table 4* soft-soft, soft-stiff and stiff-stiff. A typical three-bladed rotor operates between 0.30-0.60 Hz (Arany et al., 2017; Bhattacharya, 2019).

Table 4. The design zones

Design zones		
Soft-soft	$w < 1P$	More likely to cause resonance with the rotor
Soft-stiff	$1P < w < 3P$	Design goal
Stiff-stiff	$3P < w$	Requires a very heavy (costly) structure

$$f = \frac{\Omega}{2\pi}$$

With a maximum rotor speed of 7.6 rpm we get $f_{1P} = 0.1367$ Hz and $f_{3P} = 0.3800$ Hz.

In this analysis, a minimum clearance of 10% above the maximum rotor speed is adopted as the design target for the first natural frequency.

$$f_1 \geq 1.10 \cdot f_{1P,max}$$

With the 10% above the maximum 1P frequency we get a design target of 0.1293 Hz.

3.2 Eigenvalue and natural frequency

EoM

$$M\ddot{u} + Ku = 0$$

Eq 5

Assuming harmonic motion $u = \phi e^{i\omega t}$, Eq 5 takes the form of Eq 6

$$(K - \omega^2 M)\phi = 0$$

Eq 6

For a non-trivial solution where $\phi \neq 0$, the matrix $(K - \omega^2 M)$ must be singular, ergo

$$\det(K - \omega^2 M) = 0$$

Eq 7

For a system with matrices in the size of [24 x 24] the system becomes a degree-24 polynomial in ω^2 , which is impossible to solve analytically. This is where good programming becomes essential. However, for a 2-DOF system, the basic theory can be illustrated to later describe what the programmed function (scipy.linalg.eigh in Python, eig in Matlab) base its logic on.

$$\det \begin{vmatrix} k_{11} - \omega^2 m_{11} & k_{12} \\ k_{21} & k_{22} - \omega^2 m_{22} \end{vmatrix} = 0$$

Resulting in steps a-g

- a. $(k_{11} - \omega^2 m_{11}) \cdot (k_{22} - \omega^2 m_{22}) - k_{12} \cdot k_{21} = 0$
- b. $-k_{11} \omega^2 m_{22} - \omega^2 m_{11} k_{22} + \omega^4 m_{11} m_{22} + k_{11} k_{22} - k_{12} k_{21} = 0$
- c. $\lambda^2 m_{11} m_{22} - \lambda (k_{11} m_{22} + m_{11} k_{22}) + (k_{11} k_{22} - k_{12} k_{21}) = 0$
- d. $\lambda^2 - \lambda \left(\frac{k_{11} m_{22} + m_{11} k_{22}}{m_{11} m_{22}} \right) + \left(\frac{k_{11} k_{22} - k_{12} k_{21}}{m_{11} m_{22}} \right) = 0$
- e. $a \lambda^2 + b \lambda + c = 0$
 $a = m_{11} m_{22}$
 $b = k_{11} m_{22} + m_{11} k_{22}$
 $c = k_{11} k_{22} - k_{12} k_{21}$
- f. $\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- g. $\omega_n = \sqrt{\lambda_n} \Rightarrow f_n = \frac{\omega_n}{2\pi}$

While $\lambda = \omega^2$ gives two roots, ω has its heritage in $u(t) = \phi e^{i\omega t}$

The negative root ($-\omega$) gives the same physical oscillation as the positive ω , just describing the wave going in the other direction in the complex plane. When taking the real part (the actual physical motion) it combines into:

$$u(t) = \phi(A \cos(\omega t) + B \sin(\omega t))$$

so negative root adds no new information, it is already captured in the constants A and B.

When λ_1 and λ_2 are attained, they are substituted back into Eq 6 that now takes the format of Eq 7.

$$(K - \lambda_n M)\phi = 0$$

Eq 7

Solved for ϕ for the mode shapes since the matrix is now singular, one row redundant. This gives a ratio between two displacements, i.e., the shape. The scale here is arbitrary, which is why the mode shapes are normalized afterwards. The eigenvectors define the shape of vibration but not the amplitude, since any scalar multiple is equally valid. Mass normalization resolves this by scaling each mode shape such that the modal mass equals unity, giving a consistent basis for subsequent calculations.

$$\omega_n = 2\pi f_n$$

Eq 8

The algorithm behind the numerical method `scipy.linalg.eigh` is the Cholesky decomposition, since both K and M are symmetric positive definite, it transforms the generalized problem into a standard one with $M = LL^T$

from Eq 6 it becomes:

$$L^{-1}KL^{-T}(L^T\phi) = \omega^2(L^T\phi)$$

To then become a standard symmetric eigenvalue problem, solved efficiently by algorithms like QR iteration. It returns eigenvalues sorted ascending, so ω_1 is automatically the fundamental frequency.

3.3 Diameter Selection

The stiffness and mass of the tower are directly related by its outer diameter D_o and wall-thickness, t . To ensure that the structure is cost-efficient while keeping structural integrity, especially regarding local buckling, the structure's parameters are iteratively determined. While this could be investigated through a buckling analysis, offshore monopile standards specify a ratio in the range of 80-120 between the diameter and thickness of the cross-section (D_o/t).

The diameter has an effect on the stiffness and mass matrix via the cross-sections moment of inertia, I , and the area, A , respectively. Thus, the diameter has an effect on the structure's natural frequency as seen in Eq 5-8 in section 3.2 *Eigenvalue and natural frequency*.

The design can thereafter be tuned until the first natural frequency of the structure meets the soft-stiff target, or in this case the design target of 0.1293 Hz.

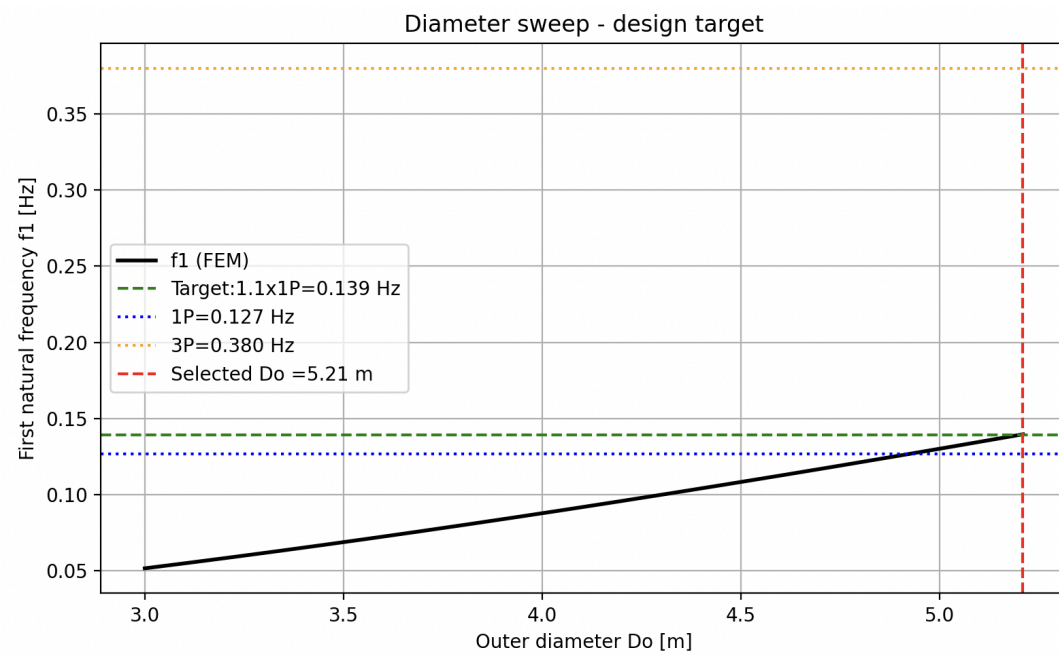
By choosing a ratio of 90, the wall thickness scales proportionally as $t = D_o/90$, keeping the cross-sectional slenderness within the required range of 80-120. The diameter is swept iteratively from 3.0 m with an addition of 0.01 m per step, until the first natural frequency meets the target.

The minimum diameter satisfying the soft-stiff requirement is found to be $D_0=5.21$ m, yielding a first natural frequency of $f_1=0.1397$ Hz, corresponding to a margin of 10.3% above the 1P frequency. The resulting cross-sectional properties are reported in *Table 5*.

Table 5, Values when diameter tuned

Notation	Value	Unit
D_o	5.210 m	m

t	0.0579 m	m
D/t	90	-
A	0.937	
I	3.11	m ⁴
f1	0.1397	Hz
f2	1.2122	Hz



3.3 Natural frequencies and mode shapes

Before adding damping to the system, an evaluation of the modeshapes are in order. The first two natural frequencies and mode shapes of the tuned structure are shown in Figure 4. Mode 1 exhibits the characteristic first bending shape of a cantilever beam, where displacement increases uniformly to a maximum at the tip. Mode 2 shows a single inflection point at approximately 80 m, where the displacement reverses direction, consistent with the second bending mode. Both shapes are normalised to unit tip displacement.

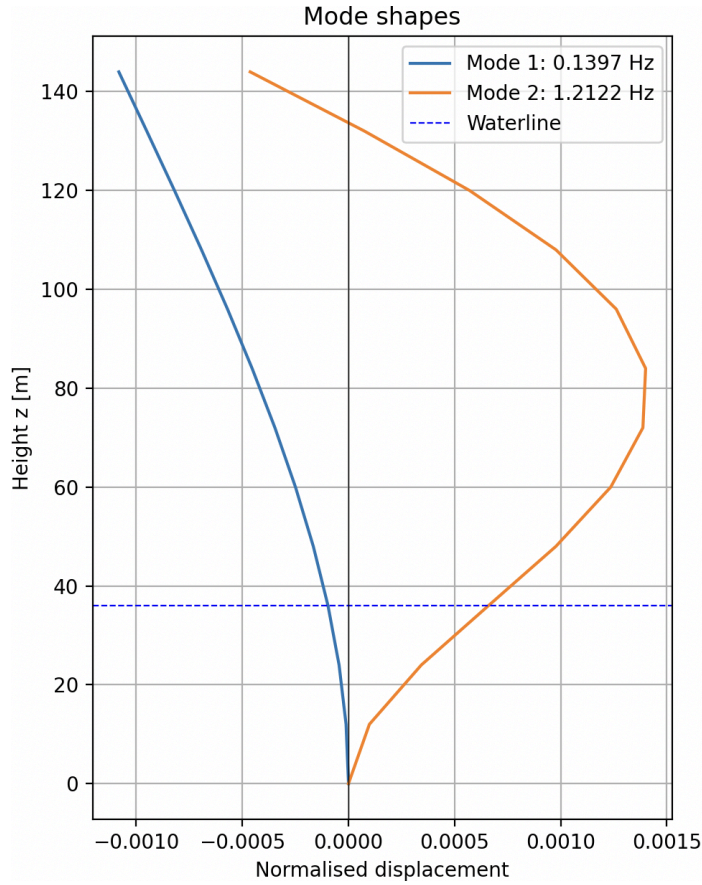


Figure 4 Mode shapes 1 & 2

For mode 1 we can see that the whole structure bends in one direction with maximum displacement at the tip and zero at the clamped base where $z=0$ is the seabed and $z=144\text{m}$ is the nacelle. This is the fundamental bending mode and the one that matters most for wind and wave loading since the excitation frequencies are closest to f_1 . For mode 2 the structure curves back on itself, creating an arch with a point of zero displacement around 80 m from base. The second mode requires much more energy to excite and is at $f_2=1.2122\text{ Hz}$, well above the wave and wind excitation frequencies for most conditions.

3.4 Damping

Since the structure dissipates energy, the system is damped. The Rayleigh damping matrix is constructed as a linear combination of the mass and stiffness matrices:

$$C = \alpha M + \beta K$$

Eq 9

$$\xi = \frac{\alpha}{2\omega} + \frac{\beta\omega}{2}$$

Eq 10

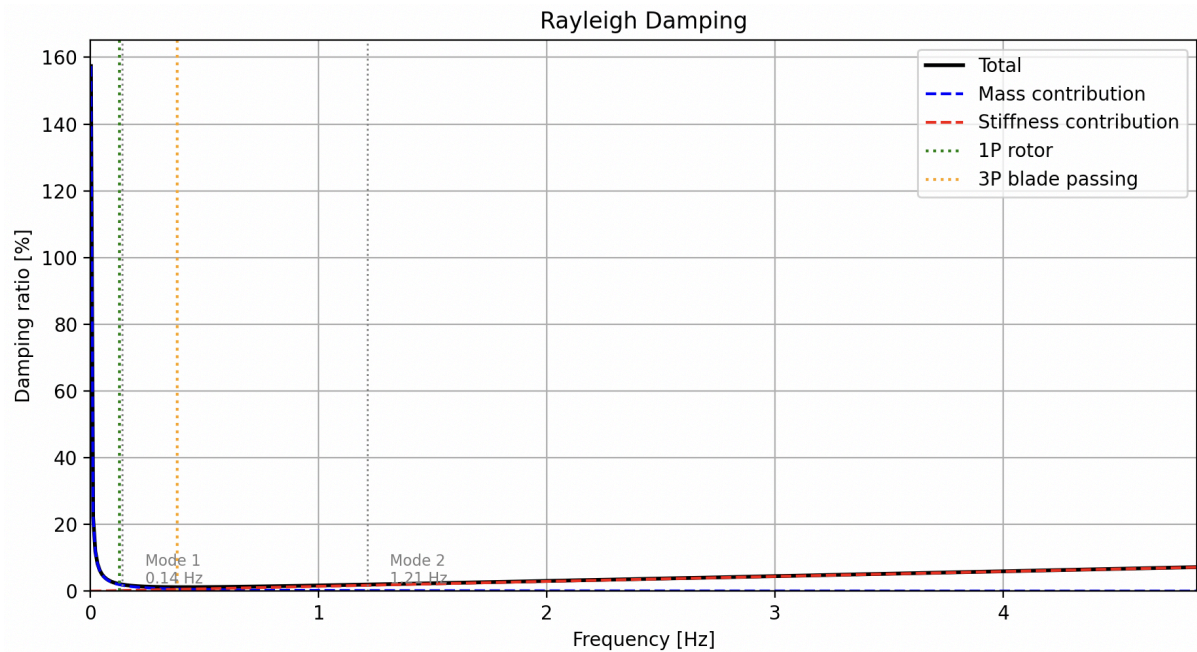
where alpha and beta are determined by prescribing 2% critical damping at both the first and second modes.

$$\frac{\alpha}{2\omega_1} + \frac{\beta\omega_1}{2} = \xi_1$$

$$\frac{\alpha}{2\omega_2} + \frac{\beta\omega_2}{2} = \xi_2$$

Solving this 2x2 system with $\xi_1 = \xi_2 = 0.02$ gives alpha=0.0315 and beta=0.0047

The Rayleigh damping curve is shown in Figure 4, with the 1P and 3P rotor frequencies marked for reference. The mass-proportional term dominates at low frequencies while the stiffness-proportional term dominates at high frequencies, with the two target modes pinned at exactly 2% damping by construction.



Although the turbine is assumed non-operating in this assignment, meaning rotor excitation is neglected in the load analysis, the frequency placement remains relevant as it governs the dynamic amplification of the structural response when excited near its natural frequency.

4. Wind & Wave Environments

Wind and wave loads act simultaneously on the structure but will in general not be perfectly aligned. In this analysis, co-directional loading in a single plane is assumed, meaning wind and wave forces are treated as collinear. This assumption does not affect the natural frequencies, which depend only on the stiffness and mass matrices, but it does influence aerodynamic damping. A load misalignment

of 90° would reduce aerodynamic damping and increase dynamic amplification, this is therefore a conservative direction for future investigation, but is not considered here.

4.1 Load Cases

Two load cases are considered, defined by different approaches to selecting the wind and wave parameters. For the loading case 1 (LC1) the worst case resonance scenario is investigated and with the loading case 2 (LC2) a high risk storm will be described. Thereby the structure is tested for both the scenario of resonance and a maximum lateral load, . The wave peak period, T_p , and the significant wave height, H_s , are used to formulate these loading cases. The fetch-wind relations are the distance over which wind blows over open water. The longer the fetch, F , the more time wind has to transfer energy into the waves. The fetch-wind relations, directly gathered from the JONSWAP spectrum, are empirical formulas, fitted to North sea measurement data by Hasselmann et al (1973).

$$T_p = \frac{0.286}{g} \left(\frac{gF}{U_{10}^2} \right)^{1/3} \cdot U_{10}$$

Eq. 11

Eq. 12 represents the wave peak period. To express loading case 1, the wave peak period is set equal to the first natural period of the structure so since the period is the inverted frequency. The corresponding reference wind speed can be expressed with Eq.12 which is derived from inverting Eq. 11 (or solved with brentq in python).

$$U_{10} = \frac{g^2}{0.0234 \cdot f_1^3 \cdot F}$$

Eq. 12

The fetch F is an implicit parameter, set to **180** km in this case. With the velocity the significant wave height can then be gathered from Eq 13. For loading case 2, LC2, this velocity is then implemented in both Eq 11 and 13 to express the wave peak period and the significant wave height, respectively. The load case parameters are presented in Table 6.

$$H_s = \frac{0.0016}{g} \sqrt{\frac{gF}{U_{10}^2}} \cdot U_{10}^2$$

Eq. 13

Table 6, parameters for the load cases LC1 & LC2.

Parameter	LC1	LC2
Fetch value F	180 km	
Reference wind speed U10	9.26 m/s	19 m/s
Wave peak period T_p	7.16 s	9.14 s
Significant wave height H_s	2.01 m	4.12 m

ω_p	0.878 rad/s	0.687 rad/s
------------	-------------	-------------

4.2 Significant Wave Heights

The significant wave heights for LC1 is $H_s=2.01$ m, corresponding to moderate sea conditions consistent with the relatively low wind speed. For LC2, $H_s=4.12$ m which reflects the stronger wind forcing. Both values are determined from the fetch-limited JONSWAP growth relations following Hasselmann et al. (1973).

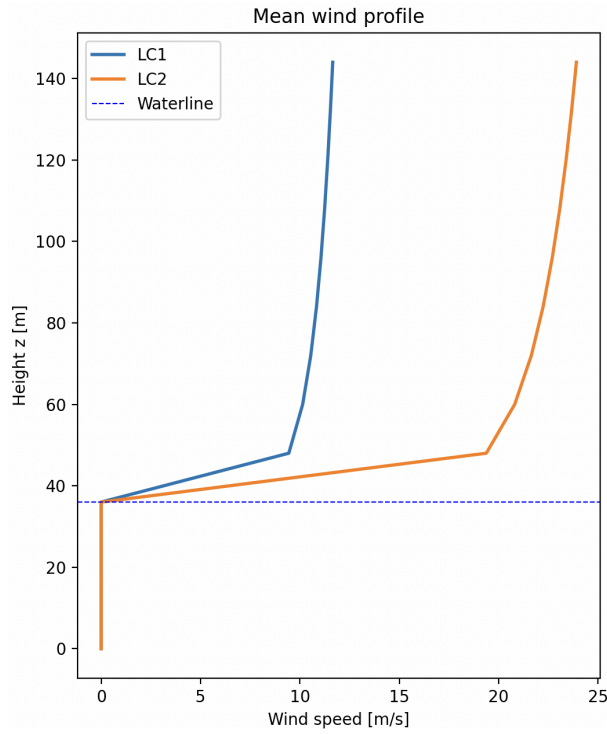
Although LC2 is characterised by significantly higher wind speeds, and thereby larger forcing, the structural response is not necessarily greater than in LC1. The wind drag force scales with the square of wind velocity, meaning LC2 produces roughly four times the wind force of LC1. However, in LC1 the wave peak period is tuned to coincide with the first natural period of the structure, concentrating wave energy at precisely the frequency the structure is most sensitive to. The resulting dynamic amplification can produce severe structural response even under moderate loading. Which load case governs the mudline moment is therefore not obvious a priori but craves further analysis.

4.3 Wind Climate, The Kaimal spectrum

The mean wind profile along the turbine height is described using a logarithmic or power law profile.

$$U(z) = U_{10} \cdot \ln\left(\frac{z}{z_0}\right) / \ln\left(\frac{10}{z_0}\right)$$

Eq.16



The turbulent wind field is characterised by the Kaimal spectrum, which defines the distribution of wind energy across frequencies at each node of the finite element model. Co-spectra are used to account for the spatial correlation of wind fluctuations between nodes. Describes how turbulent wind energy is distributed across frequencies.

$$S_u(f, z) = \frac{4 \cdot \sigma_u^2 \cdot \frac{L_u}{U(z)}}{(1 + 6 \cdot f \cdot \frac{L_u}{U(z)})^{5/3}}$$

Eq 14

Where $\sigma_u = I_u \cdot U$ is the turbulence standard deviation and L_u is the length scale. The 5/3 exponent comes from Kolmogorov's turbulence theory telling that energy cascades from large slow waves down to small fast ones following the power law, ergo, the energy is intact it just takes another form. The co-spectrum between two nodes at two heights accounts for the fact that nearby points see correlated turbulence, typically written as in Eq 13, where the coherence decays exponentially with separation distance and frequency.

$$S_u(f, z_i, z_j) = \sqrt{S_u(f, z_i) \cdot S_u(f, z_j)} \cdot coherence(f, z_i, z_j)$$

Eq 15

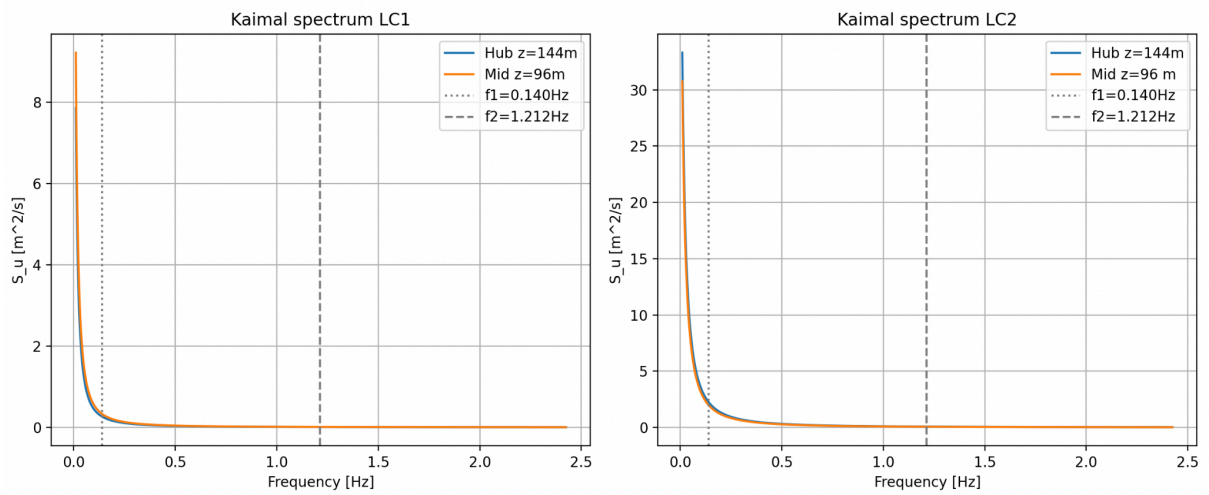


Figure 5, The Kaimal spectrums for loading case 1 and 2

The Kaimal spectra for both load cases are shown in *Figure 5*. The spectra peak at very low frequencies, reflecting the dominance of large-scale turbulent eddies that carry most of the wind energy. Energy decays rapidly with increasing frequency following Kolmogorov's -5/3 power law. By the first natural frequency $f_1=0.1397$ Hz, the spectral density has already decayed significantly, to approximately $0.1 \text{ m}^2/\text{s}$ for LC1 and $0.25 \text{ M}^2/\text{s}$ for LC2 at hub height. This indicated that direct wind excitation at the structural natural frequency is relatively modest, and that dynamic amplification is more likely to be driven by wave loading concentrated near f_1 , particularly in LC1 where the wave peak period is tuned to the natural period.

4.4 Wave Climate

The wave climate is described using the JONSWAP spectrum, appropriate for fetch-limited seas such as the North Sea. The spectrum is parameterized by the significant wave height H_s and peak period T_p , both derived from the fetch wind relations for each load case. The JONSWAP spectral density is:

$$S(\omega) = \frac{\alpha g^2}{\omega^5} e^{a1}$$

$$a1 = \left(-\beta \left(\frac{\omega_p}{\omega} \right)^4 \right) \cdot \gamma^{a2}$$

$$a2 = e^{-a3}$$

$$a3 = \frac{(\omega - \omega_p)^2}{2\omega_p^2 \sigma^2}$$

$$\beta = \frac{5}{4}$$

ω is wave frequency while ω_p is peak wave-frequency, $\gamma = 3.3$ is the peak enhancement factor giving JONSWAP its characteristic narrow peak compared to open-ocean spectra, and $\sigma = 0.07$ for $\omega \leq \omega_p$, $\sigma = 0.09$ for $\omega > \omega_p$. The parameter α is calibrated iteratively to match the target H_s .

beta is a shape factor with a default value 1.25. As shown in Figure 6, the LC1 spectrum peaks at ω

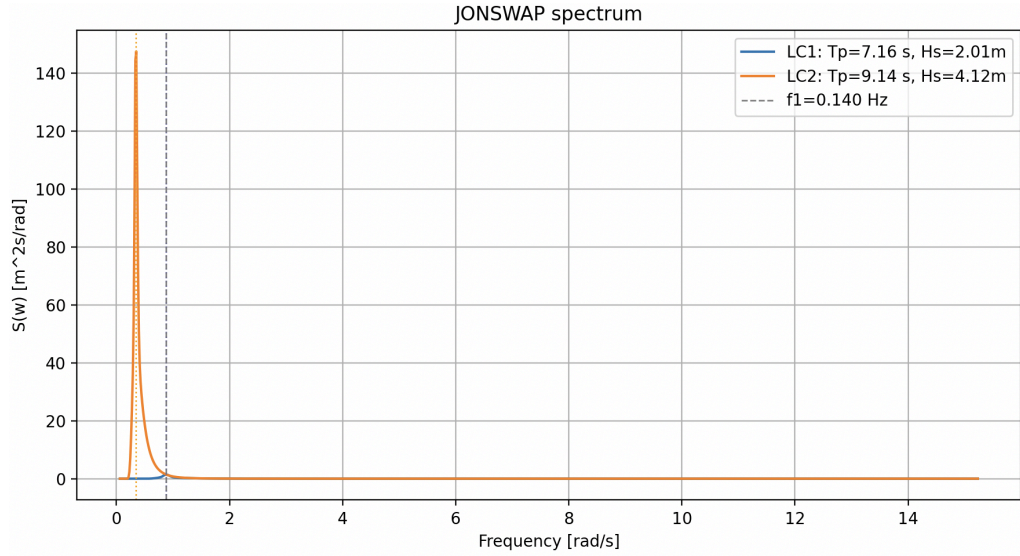


Figure 6

5. Time Series Generation

5.1 Synthesis Methodology

Time series of wind and wave kinematics are generated from their respective spectra using the random phase angle method. For each node of the finite element model, a discretised frequency axis is defined and the amplitude of each frequency component is determined from the spectral density. A random phase angle, drawn from a uniform distribution between 0 and 2π , is assigned to each component. The time series is then constructed as a superposition of harmonic components, yielding a stationary Gaussian process consistent with the target spectrum.

The Morison drag term is:

$$F_{drag} = C_d \frac{D}{2} \cdot (u - \dot{x})|u - \dot{x}| \quad \text{Eq (xx)}$$

which is non-linear because of the absolute value, and coupled because it contains $\dot{x}$. By dropping the $\dot{x}$ -term the term gets linearised.

$$F_{drag} = C_d \frac{D}{2} \cdot u|u| \quad \text{Eq (xx)}$$

Making the force only depends on the fluid velocity, not the structural response. The system is thereby decoupled and we can thereby work in the frequency domain using the transfer function. However this strategy gives us merely an approximation.

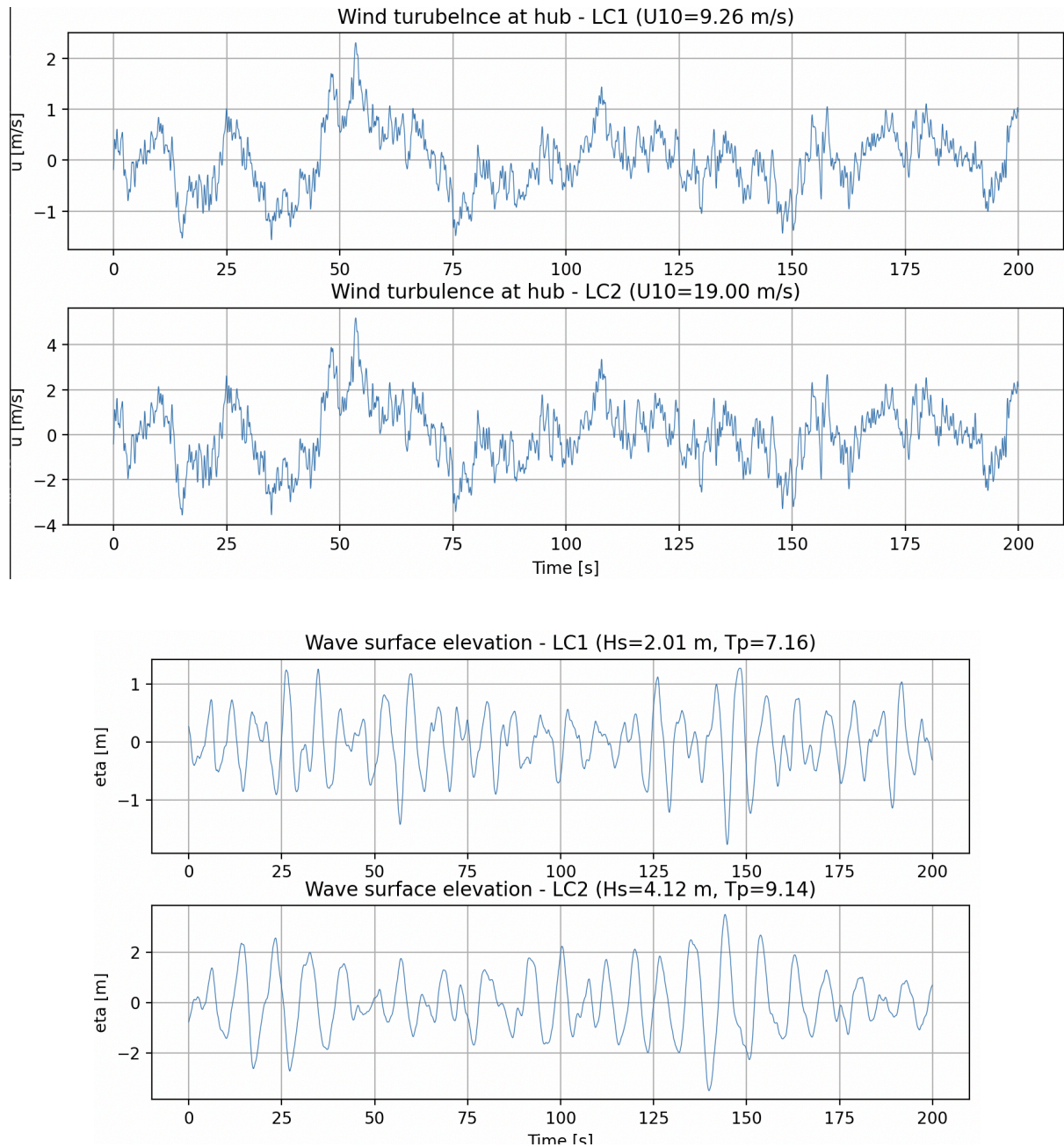
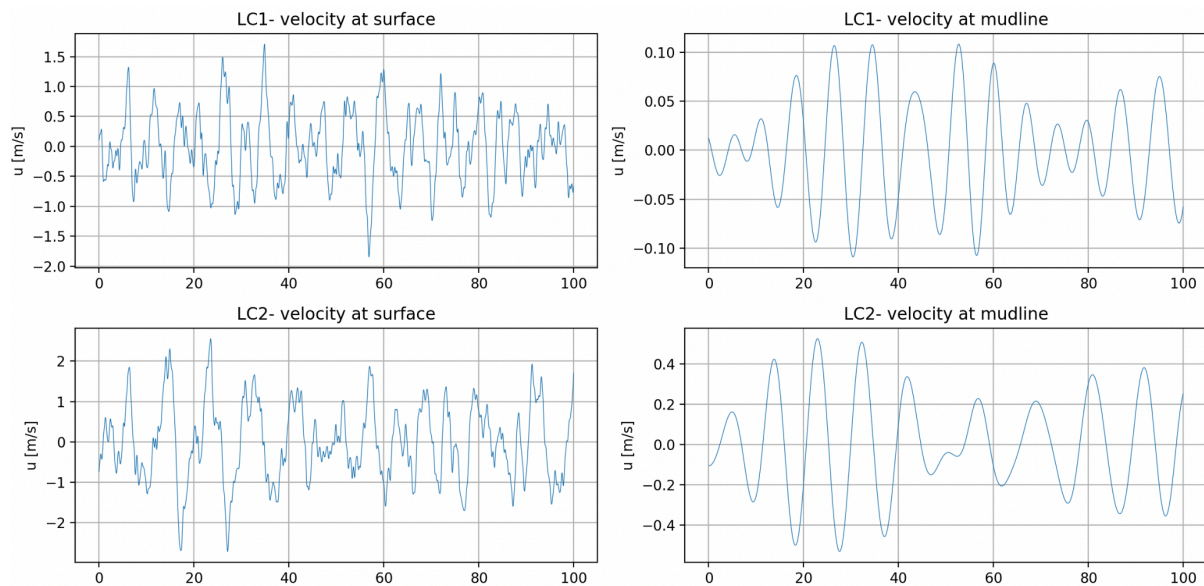


Figure 8

The resulting surface elevation time series are shown in Figure 8. The irregular character of the wave field is immediately visible, individual wave heights and periods vary continuously, consistent with a real ocean sea state. For LC1 the surface elevation ranges approximately between -1 and +1 m, reflecting the modest significant wave heights of $H_s=2$ m. For LC2 the amplitude roughly doubles to 4.12 m. The dominant period is also visible, waves in LC1 cycle approximately every 7 seconds while LC2 waves are slightly slower and broader at approximately 9 seconds, consistent with the respective peak periods $T_p=7.16$ s and $T_p=9.14$ s. Occasional waves noticeably larger than the surrounding sea surface are also visible, reflecting the statistical variability inherent in a random sea state. These large individual waves arise naturally from the superposition of many harmonic components and represent the tail of the wave height distribution. A regular wave analysis using only the peak period would miss the broadband nature of the loading and potentially underestimate fatigue damage from the full frequency content. The surface elevation is subsequently converted to water particle velocities

and accelerations at each submerged node using linear wave theory, which then serve as input to the Morison equation to determine the hydrodynamic loading on the structure.



- Random phase angle method
- harmonic superposition
- stationary gaussian process
- drag linearisation
- decoupling of forcing and response

5.2 Series Length and Convergence

The duration of the time series must be sufficiently long to ensure stable estimates of the statistical properties of the response, such as the standard deviation and mean zero-crossing period. A convergence analysis is performed by progressively increasing the series length and monitoring the convergence of these quantities. The final duration is selected where further increase no longer produces significant changes in the statistics.

A convergence analysis is performed by progressively increasing the time series duration from 100 to 1800 seconds and monitoring the standard deviation and mean zero crossing period of the surface elevation. The results are shown in Figure 10. Both quantities show initial fluctuations that gradually stabilise with increasing duration. The standard deviation of LC1 is the last to converge, showing a damped oscillation that flattens by approximately 1750 seconds. A total duration of 1800 seconds is therefore adopted for all analyses, providing sufficient statistical stability for both load cases.

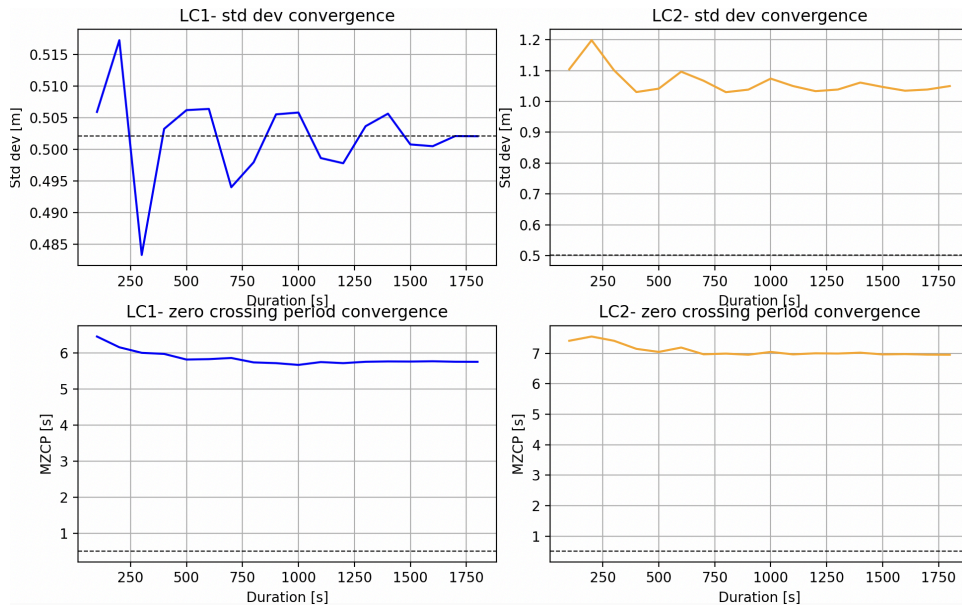


Figure 10

- Convergence analysis
- Standard deviation
- Mean zero-crossing period
- Time series duration (1800 s)

5.3 Sampling frequency

The sampling frequency is chosen to adequately resolve the first and second natural frequencies of the structure. Following the Nyquist criterion, the sampling frequency must be at least twice the highest frequency of interest. In practice a margin above this minimum is adopted to avoid aliasing and to accurately capture the dynamic response.

State actual sampling frequencies used, show it satisfies nyquist, note frequency resolution $\Delta f = 1/T$

6. Frequency Domain Analysis (Linear)

6.1 Linearisation of Forces

The frequency domain analysis is based on a linearised formulation of the hydrodynamic and aerodynamic forces. Linearisation is achieved by discarding the contribution of the structural response from the forcing, meaning the relative velocity between the fluid and structure is replaced by the fluid velocity alone. This decouples the forcing from the structural motion and allows the analysis to be carried out in the frequency domain. The assumption is valid when the structural response is small relative to the fluid kinematics.

A more refined approach is statistical linearisation, in which the non-linear drag term is replaced by an equivalent linear damping coefficient C_{eq} , determined by minimising the mean square error between the original and linearised force. Since C_{eq} depends on the standard deviation of the relative velocity, the procedure is iterative, an initial response estimate is used to compute C_{eq} , the system is solved, and the coefficient is updated until convergence. This approach remains in the frequency domain while partially accounting for non-linear drag, but is not pursued here.

- Linearised drag
- Fluid velocity assumption
- Decoupling
- Statistical linearisation (C_{eq})

6.2 Force Spectra

Based on the linearised force formulation, the power spectral density of the wind and wave loads is determined at each node of the finite element model. The wind force spectrum is derived from the Kaimal spectrum and the co-spectra accounting for spatial correlation along the turbine height. The wave force spectrum follows from the JONSWAP spectrum through the linearised Morison equation.

$$S_{FF}(\omega) = |H_{Morison}(\omega)|^2 \cdot S_{\eta\eta}(\omega)$$

$$H_{Morison}(\omega) = \rho C_m \frac{\pi D^2}{4} \cdot (-\omega^2) \cdot \frac{\cosh(k(z+h))}{\cosh(kh)}$$

- Wind force PSD
- Wave force PSD
- nodal force distribution
- spation correlation

6.3 Mudline Moment

The one-sided power spectral density of the bending moment at the mudline is obtained through a frequency domain transfer from the nodal force spectra. The frequency response function of the finite element model relates the input force spectrum at each node to the resulting moment at the mudline. Standard deviations and other statistical quantities are subsequently derived by integration of the moment spectrum.

$$S_{MM}(\omega) = |H(\omega)|^2 \cdot S_{FF}(\omega)$$

$$\sigma_M^2 = \int_0^{\infty} S_{MM}(\omega) d\omega$$

Numerical integration using the trapezoidal rule was applied over the discrete frequency axis.

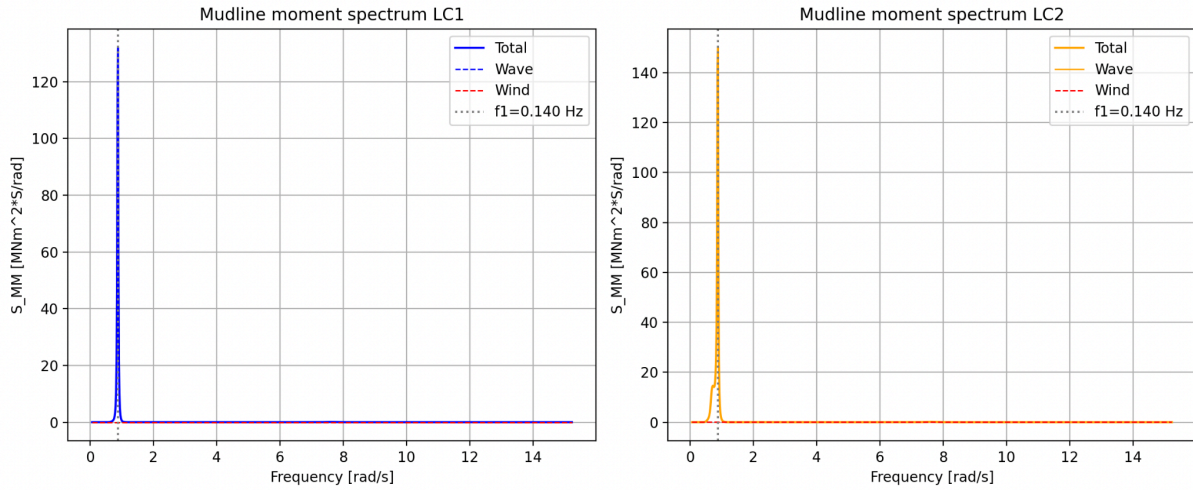


Figure 11, power spectral density at mudline

The one-sided power spectral density of the mudline moment is shown in figure 11 for both load cases. Both spectra show a sharp peak at the first natural frequency $f_1 = 0.1397$ Hz ($\omega = 0.878$ rad/s), reflecting the structural amplification at resonance. For LC2 a secondary feature is visible at lower frequencies around $\omega = 0.69$ rad/s, corresponding to the JONSWAP peak period $T_p = 9.14$ s. The wind contribution is negligible in both load cases, confirming that the mudline moment is dominated entirely by wave loading at these frequencies.

Despite LC1 being designed with the wave peak period tuned to the structural natural period, the peak spectral density at f_1 is larger for LC2 ($150 \text{ MNm}^2\text{s/rad}$) than for LC1 ($132 \text{ MNm}^2\text{s/rad}$). This reflects the significantly larger wave height of LC2 ($H_s = 4.12$ vs 2.01 m), the total wave energy in LC2 more than compensates for the lack of perfect frequency alignment. The standard deviation of the mudline moment is 2.601 MNm for LC1 and 3.308 MNm for LC2.

	LC1	LC2
peak PSD	$132 \text{ MNm}^2\text{s}$	$150 \text{ MNm}^2\text{s/rad}$
σ	2.601 MNm	3.308 MNm

- frequency response function $H(w)$

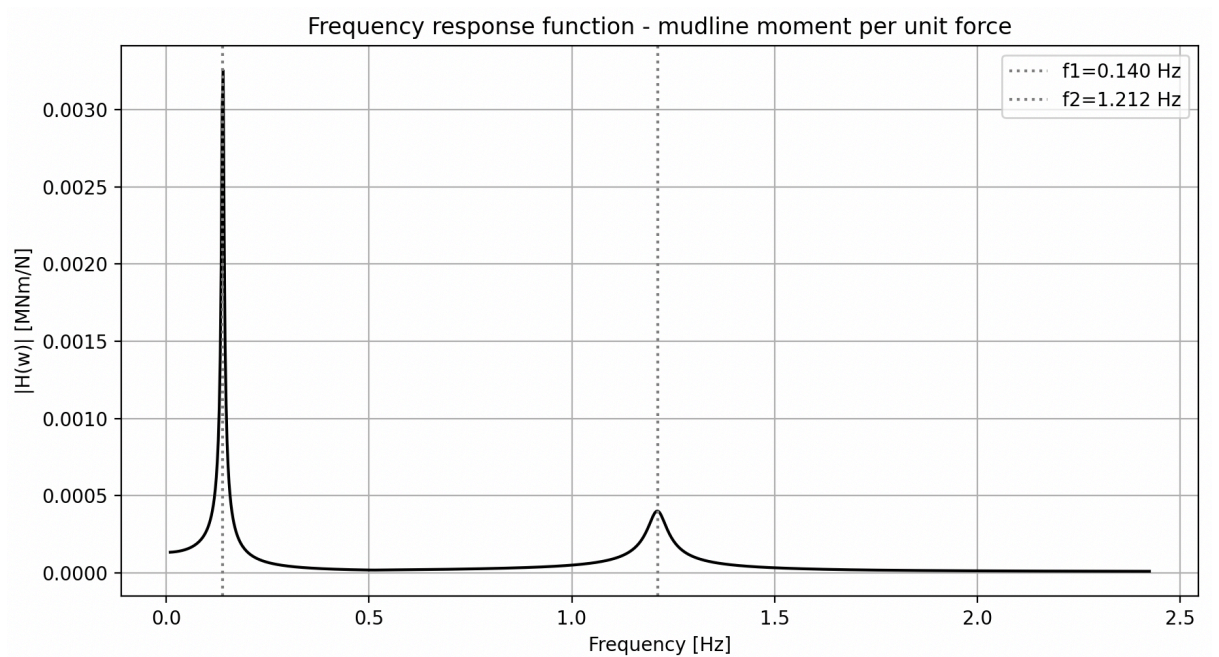
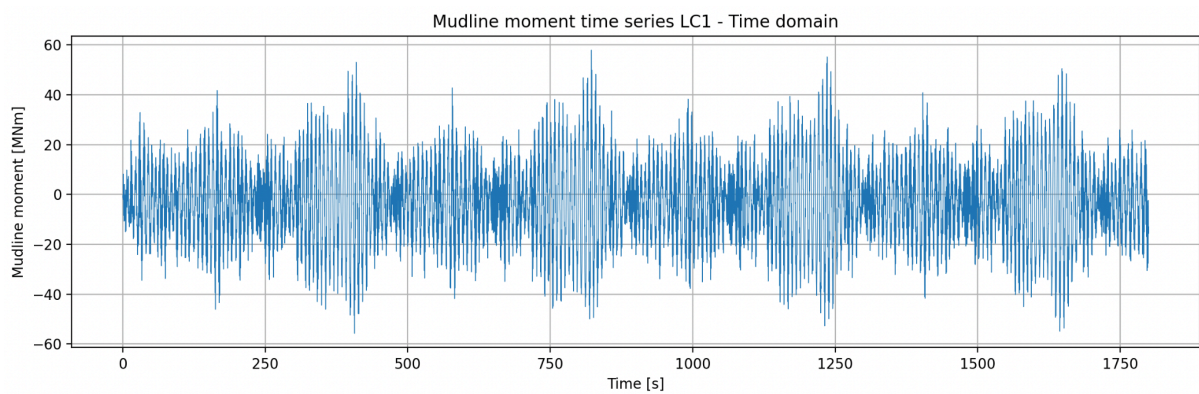


Figure 12

In figure 12 we see the frequency response function of the structure relating unit nodal force to mudline bending moment. Two resonance peaks are visible at $f_1 = 0.1397$ Hz and $f_2 = 1.2122$ Hz, the first mode dominating. The quasi-static baseline of 0.00015 MNm/N reflects the static flexibility of the structure.

	LC1	LC2
std mudline moment	2.601 MNm	3.308 MNm
mean zero crossing period T_z	6.00 s	6.17 s
S_{MM} peak at f_1	132 MNm ² /rad	150 MNm ² /rad

- Transfer function



The probability of exceedance

$$P(|x| > x_{crit}) = 1 - \operatorname{erf}\left(\frac{x_{crit}}{\sigma\sqrt{2}}\right)$$

Results from code is giving an exploded mudline moment

- The frequency domain result of 2.6 MNm is the linearised response to wave loading only, using spectral methods. The time domain result of 15.9 MNm includes the full nonlinear Morison equation with relative kinematics, mean wind forces, and turbulent wind. These are genuinely different models and some discrepancy is expected and physically meaningful, the whole point of step 7, comparing. A factor of 6 is larger than typical but not impossible given the strongly nonlinear drag term for a 5.21 m diameter pile.

- Frequency domain force spectrum may underestimate forces because it uses a simplified linearisation that does not capture the full inertia contribution over the element length
- The time domain may overestimate because the mean wind creates a large static response that interacts with the dynamic wave loading through the nonlinear drag term
- The factor of 6 may partly reflect the genuine difference between linearised frequency domain and fully nonlinear time domain, this is physically expected to some degree and is the core topic of step 7 (chp 8)

See script for full list of diagnostics that have been made to find the issue

7. Time Domain Analysis (Non-linear)

$$dF = C_m \cdot \frac{\pi D^4}{4} \cdot (\ddot{u} - \ddot{x}) + C_d \cdot \frac{D}{2} \cdot (u - \dot{x})|u - \dot{x}|$$

- Non-linear morison equation
- Relative velocity
- Coupled forcing and response

8. Comparison of Results

9. Diffraction Assessment

$$\lambda = \frac{2\pi}{k}$$

$$\text{where } k \text{ satisfies } \omega^2 = gk \cdot \tan(kh)$$

At the peak frequency of LC2, check whether:

$$\text{if } D < 0.2\lambda \Rightarrow \text{Morison valid}$$

- Morison validity condition
- $D < 0.2$ lambda criterion
- slender structure assumption

10. Conclusions

11. References

Appendices